Hyperplanes, linear classification and Fisher's LDA

MATH 2740 – Mathematics of Data Science – Unit U06

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The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

Outline

Supervised vs unsupervised linear methods

Geometry of hyperplanes and support vector machines

Least-squares classification

Fisher's linear discriminant analysis (LDA)

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Unsupervised vs supervised linear methods

In Unit 05 (PCA), we studied **unsupervised** dimensionality reduction:

- Given unlabelled feature vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$
- Find orthogonal directions that maximise total sample variance

Here, we study **supervised binary classification**:

- Given a labelled training dataset of n observations

$$(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)$$

  where $\boldsymbol{x}_i \in \mathbb{R}^d$ is a feature vector and $y_i \in \{-1, +1\}$ (or $\{1, 2\}$) indicates the class to which $\boldsymbol{x}_i$ belongs
- Find a linear decision boundary or 1D projection that separates the two classes and predicts the label $\hat{y}(\boldsymbol{x})$ of any new point $\boldsymbol{x} \in \mathbb{R}^d$

Does maximum variance always separate classes?

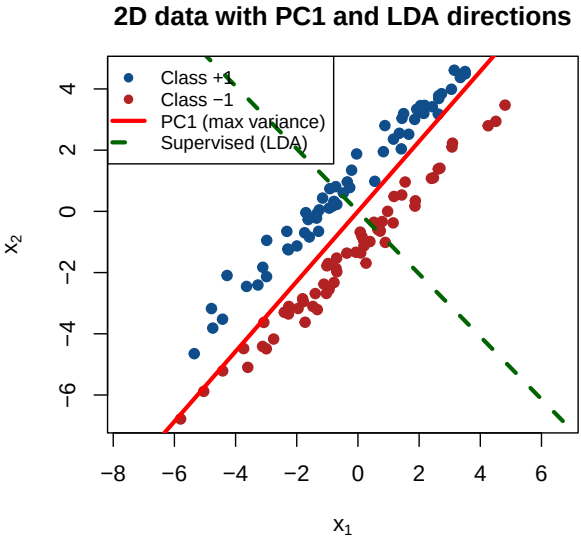
Suppose two classes form elongated, parallel point clouds that are closely spaced along their perpendicular direction

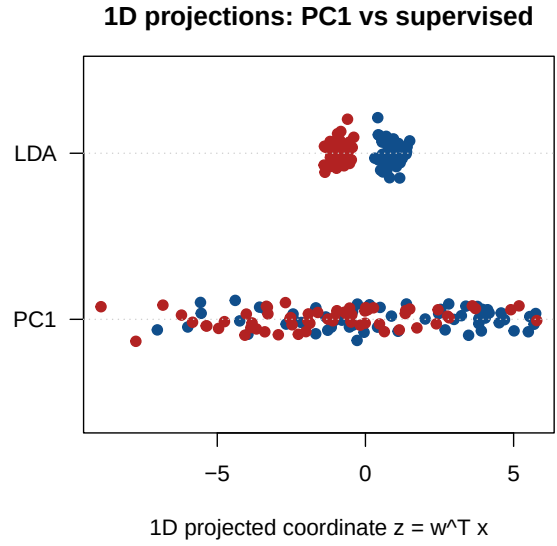
- Because PCA ignores the class labels y_i, it pools all n points together and finds the direction of largest overall spread
- When the spread *along* the clouds is larger than the distance *between* the clouds, the first principal component (PC1) runs parallel to the clouds
- Projecting onto PC1 mixes the two classes completely, whereas projecting onto the perpendicular direction separates them cleanly

When PCA fails to separate classes

Simulate two elongated parallel classes in $\mathbb{R}^2$ and compare PC1 (red) with the supervised separating direction (green)

```
set.seed(2740)
n_c = 60
t1 = rnorm(n_c, 0, 2.5); t2 = rnorm(n_c, 0, 2.5)
X1 = cbind(t1 + rnorm(n_c, 0, 0.25), t1 + 1.2 + rnorm(n_c, 0, 0.25))
X2 = cbind(t2 + rnorm(n_c, 0, 0.25), t2 - 1.2 + rnorm(n_c, 0, 0.25))
X_all = rbind(X1, X2)
y_all = c(rep(1, n_c), rep(-1, n_c))
pc1 = eigen(cov(X_all))$vectors[, 1]
SW = (n_c - 1) * (cov(X1) + cov(X2))
w_lda = solve(SW, colMeans(X1) - colMeans(X2))
w_lda = w_lda / sqrt(sum(w_lda^2))
```


2D data with PC1 and LDA directions
Class +1
Class −1
PC1 (max variance)
Supervised (LDA)
x2
x1
4
2
0
−2
−4
−6
−8
−6
−4
−2
0
2
4
6


1D projections: PC1 vs supervised
LDA
PC1
−5
0
5
1D projected coordinate z = w^T x

Three linear classification frameworks in this unit

All three methods use a linear score $f(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{x} + b$ to classify points, but choose $\boldsymbol{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$ using different linear algebraic principles:

1. **Geometry of hyperplanes and support vector machines (SVM)**: characterise decision boundaries via normal vectors, orthogonal projections and maximum-margin separation

2. **Least-squares classification**: treat binary labels $y_i \in \{-1, +1\}$ as regression targets and solve the normal equations $(X^TX)\tilde{\boldsymbol{w}} = X^T\boldsymbol{y}$ from Unit 03

3. **Fisher's linear discriminant analysis (LDA)**: project data onto a line $\boldsymbol{w}$ that maximises between-class separation relative to within-class scatter via the generalised Rayleigh quotient

Supervised vs unsupervised linear methods

Geometry of hyperplanes and support vector machines

Least-squares classification

Fisher's linear discriminant analysis (LDA)

Geometry of hyperplanes and support vector machines

Hyperplanes and normal vectors

Orthogonal distance to a hyperplane

Linear separability and hard-margin SVM

Hyperplanes in $\mathbb{R}^d$

Definition 88 (Hyperplane in $\mathbb{R}^d$)

*Let $\boldsymbol{w} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$ be a non-zero vector and $b \in \mathbb{R}$ be a scalar. The **hyperplane** determined by $\boldsymbol{w}$ and b is the set of points*

$$\mathcal{H} = \left\{ \boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T \boldsymbol{x} + b = 0 \right\} = \left\{ \boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w} \bullet \boldsymbol{x} + b = 0 \right\}$$

*The vector $\boldsymbol{w}$ is called a **normal vector** to $\mathcal{H}$ and b is called the **offset** (or **bias**)*

- In $\mathbb{R}^2$, a hyperplane is a line $w_1 x_1 + w_2 x_2 + b = 0$
- In $\mathbb{R}^3$, a hyperplane is a plane $w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$
- Multiplying $\boldsymbol{w}$ and b by any non-zero constant $\alpha \neq 0$ leaves the set $\mathcal{H}$ unchanged

Linear subspace vs affine subspace

When is a hyperplane $\mathcal{H} = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T\boldsymbol{x} + b = 0\}$ a vector subspace of $\mathbb{R}^d$?

- A vector subspace must contain the zero vector $\mathbf{0}$, and $\mathbf{0} \in \mathcal{H} \iff \boldsymbol{w}^T\mathbf{0} + b = 0 \iff b = 0$
- When $b = 0$, the set $\mathcal{H}_0 = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T\boldsymbol{x} = 0\} = \{\boldsymbol{w}\}^\perp$ is the orthogonal complement of span($\boldsymbol{w}$), which is a $(d-1)$-dimensional vector subspace of $\mathbb{R}^d$
- When $b \neq 0$, $\mathcal{H}$ does not pass through the origin; it is an **affine subspace** obtained by translating $\mathcal{H}_0$ by any fixed point $\boldsymbol{x}^* \in \mathcal{H}$:

$$\mathcal{H} = \boldsymbol{x}^* + \mathcal{H}_0 = \left\{\boldsymbol{x}^* + \boldsymbol{u} : \boldsymbol{w}^T\boldsymbol{u} = 0\right\}$$

Example in $\mathbb{R}^2$ – Hyperplanes with $b = 0$ and $b \neq 0$

In $\mathbb{R}^2$, hyperplanes are lines orthogonal to $\boldsymbol{w} \neq \mathbf{0}$

- **Line through the origin ($b = 0$):**

  $$\mathcal{H}_0 = \left\{ \boldsymbol{x} \in \mathbb{R}^2 : \boldsymbol{w}^T \boldsymbol{x} = 0 \right\} = \{\boldsymbol{w}\}^\perp$$

  Since $\mathbf{0} \in \mathcal{H}_0$, it is a 1D **vector subspace**

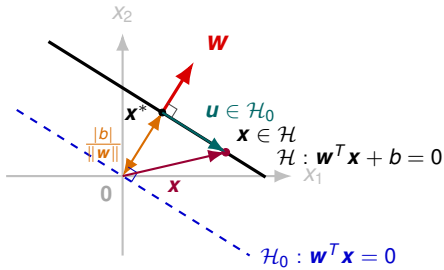
- **Line not through origin ($b \neq 0$):**

  $$\mathcal{H} = \left\{ \boldsymbol{x} \in \mathbb{R}^2 : \boldsymbol{w}^T \boldsymbol{x} + b = 0 \right\}$$

  $\mathbf{0} \notin \mathcal{H}$; **affine subspace** translated by $\boldsymbol{x}^* \in \mathcal{H}$:

  $$\mathcal{H} = \boldsymbol{x}^* + \mathcal{H}_0 = \{\boldsymbol{x}^* + \boldsymbol{u} : \boldsymbol{u} \in \mathcal{H}_0\}$$

- Every $\boldsymbol{x} \in \mathcal{H}$ satisfies $\boldsymbol{w}^T \boldsymbol{x} = -b$



Why $\boldsymbol{w}$ is orthogonal to the hyperplane

Proposition 89 (Orthogonality of the normal vector)

Let $\mathcal{H} = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T\boldsymbol{x} + b = 0\}$. For any two points $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{H}$, the displacement vector $\boldsymbol{x}_1 - \boldsymbol{x}_2$ lying along $\mathcal{H}$ is orthogonal to $\boldsymbol{w}$:

$$\boldsymbol{w} \bullet (\boldsymbol{x}_1 - \boldsymbol{x}_2) = \boldsymbol{w}^T(\boldsymbol{x}_1 - \boldsymbol{x}_2) = 0$$

Proof:

- Since $\boldsymbol{x}_1 \in \mathcal{H}$ and $\boldsymbol{x}_2 \in \mathcal{H}$, both points satisfy the hyperplane equation:

$$\boldsymbol{w}^T\boldsymbol{x}_1 + b = 0 \qquad \text{and} \qquad \boldsymbol{w}^T\boldsymbol{x}_2 + b = 0$$

- Subtracting the second equation from the first eliminates b:

$$(\boldsymbol{w}^T\boldsymbol{x}_1 + b) - (\boldsymbol{w}^T\boldsymbol{x}_2 + b) = \boldsymbol{w}^T(\boldsymbol{x}_1 - \boldsymbol{x}_2) = 0$$

- Hence $\boldsymbol{w} \perp (\boldsymbol{x}_1 - \boldsymbol{x}_2)$ for every pair of points $\boldsymbol{x}_1, \boldsymbol{x}_2$ on $\mathcal{H}$ □

Point-normal form of a (hyper)plane

A hyperplane is determined by a **point** $\boldsymbol{x}_0 \in \mathcal{H}$ and a **normal vector** $\boldsymbol{w} \neq \mathbf{0}$

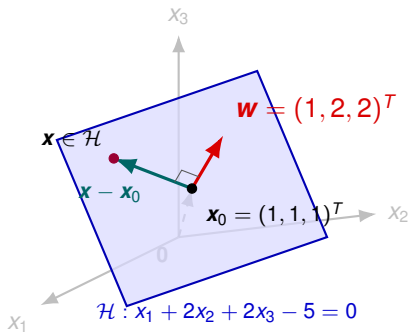
- For any $\boldsymbol{x} \in \mathcal{H}$, the displacement $\boldsymbol{x} - \boldsymbol{x}_0$ lies along $\mathcal{H}$, so $\boldsymbol{w} \perp (\boldsymbol{x} - \boldsymbol{x}_0)$:

  $\boldsymbol{w}^T(\boldsymbol{x} - \boldsymbol{x}_0) = 0$ (**point-normal form**)
- Expanding gives $\boldsymbol{w}^T\boldsymbol{x} - \boldsymbol{w}^T\boldsymbol{x}_0 = 0$, which is

  $\boldsymbol{w}^T\boldsymbol{x} + b = 0$ with $b = -\boldsymbol{w}^T\boldsymbol{x}_0$
- **Example in $\mathbb{R}^3$:** Point $\boldsymbol{x}_0 = (1, 1, 1)^T$ and normal $\boldsymbol{w} = (1, 2, 2)^T$:

  $$1(x_1 - 1) + 2(x_2 - 1) + 2(x_3 - 1) = 0$$
  $$\iff x_1 + 2x_2 + 2x_3 - 5 = 0$$

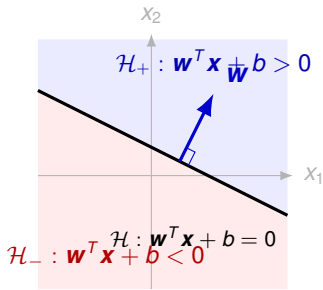
  Here $b = -\boldsymbol{w}^T\boldsymbol{x}_0 = -(1 + 2 + 2) = -5$



Half-spaces and linear classifiers

A hyperplane $\mathcal{H}$ divides $\mathbb{R}^d \setminus \mathcal{H}$ into two open **half-spaces**:

$$\mathcal{H}_+ = \left\{ \boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T \boldsymbol{x} + b > 0 \right\}$$
$$\mathcal{H}_- = \left\{ \boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T \boldsymbol{x} + b < 0 \right\}$$



- The normal vector $\boldsymbol{w}$ points into the positive half-space $\mathcal{H}_+$
- The corresponding **linear classifier** assigns class labels via the sign function:

$$\hat{y}(\boldsymbol{x}) = \text{sgn}(\boldsymbol{w}^T \boldsymbol{x} + b) = \begin{cases} +1 & \text{if } \boldsymbol{x} \in \mathcal{H}_+ \\ -1 & \text{if } \boldsymbol{x} \in \mathcal{H}_- \end{cases}$$

Example – Hyperplane geometry and classification in $\mathbb{R}^2$

Using the line $\mathcal{H}$ in $\mathbb{R}^2$ given by $2x_1 - x_2 - 3 = 0$, classify $\boldsymbol{p} = (3,1)^T$ and $\boldsymbol{q} = (0,0)^T$

- **Normal vector and offset:** $\boldsymbol{w} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $b = -3$, with norm $\|\boldsymbol{w}\| = \sqrt{2^2 + (-1)^2} = \sqrt{5}$
- **Two points on $\mathcal{H}$:** setting $x_1 = 0$ gives $\boldsymbol{x}_1 = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$; setting $x_2 = 1$ gives $\boldsymbol{x}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$
- **Orthogonality check:** $\boldsymbol{x}_2 - \boldsymbol{x}_1 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ and $\boldsymbol{w}^T(\boldsymbol{x}_2 - \boldsymbol{x}_1) = 2(2) + (-1)(4) = 0$
- **Classifying test points with $\hat{y}(\boldsymbol{x}) = \text{sgn}(\boldsymbol{w}^T\boldsymbol{x} + b)$:**
  - For $\boldsymbol{p} = (3,1)^T$: $\boldsymbol{w}^T\boldsymbol{p} + b = 2(3) - 1(1) - 3 = +2 > 0 \implies \hat{y}(\boldsymbol{p}) = +1$ ($\boldsymbol{p} \in \mathcal{H}_+$)
  - For $\boldsymbol{q} = (0,0)^T$: $\boldsymbol{w}^T\boldsymbol{q} + b = 2(0) - 1(0) - 3 = -3 < 0 \implies \hat{y}(\boldsymbol{q}) = -1$ ($\boldsymbol{q} \in \mathcal{H}_-$)

Geometry of hyperplanes and support vector machines

Hyperplanes and normal vectors

Orthogonal distance to a hyperplane

Linear separability and hard-margin SVM

Orthogonal distance from a point to a hyperplane

Theorem 90 (Distance from a point to a hyperplane)

Let $\mathcal{H} = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T\boldsymbol{x} + b = 0\}$ be a hyperplane with normal vector $\boldsymbol{w} \neq \boldsymbol{0}$, and let $\boldsymbol{x}_0 \in \mathbb{R}^d$ be any point. The Euclidean distance from $\boldsymbol{x}_0$ to $\mathcal{H}$ is

$$d(\boldsymbol{x}_0, \mathcal{H}) = \min_{\boldsymbol{x} \in \mathcal{H}} \|\boldsymbol{x}_0 - \boldsymbol{x}\| = \frac{|\boldsymbol{w}^T\boldsymbol{x}_0 + b|}{\|\boldsymbol{w}\|}$$

- Setting $\boldsymbol{x}_0 = \boldsymbol{0}$ gives the orthogonal distance from the origin to $\mathcal{H}$:

$$d(\boldsymbol{0}, \mathcal{H}) = \frac{|b|}{\|\boldsymbol{w}\|}$$

- The quantity $\frac{\boldsymbol{w}^T\boldsymbol{x}_0 + b}{\|\boldsymbol{w}\|}$ is the **signed distance** from $\boldsymbol{x}_0$ to $\mathcal{H}$ (positive on $\mathcal{H}_+$, negative on $\mathcal{H}_-$)

Proof – Distance from a point to a hyperplane

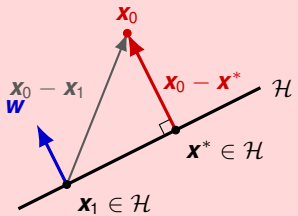
Let $\boldsymbol{x}^* \in \mathcal{H}$ be the orthogonal projection of $\boldsymbol{x}_0$ onto $\mathcal{H}$, and pick any $\boldsymbol{x}_1 \in \mathcal{H}$

- Since $\boldsymbol{x}_0 - \boldsymbol{x}^* \perp \mathcal{H}$, it equals the orthogonal projection of $\boldsymbol{x}_0 - \boldsymbol{x}_1$ onto $\boldsymbol{w}$:

$$\boldsymbol{x}_0 - \boldsymbol{x}^* = \text{proj}_{\boldsymbol{w}}(\boldsymbol{x}_0 - \boldsymbol{x}_1) = \frac{\boldsymbol{w}^T(\boldsymbol{x}_0 - \boldsymbol{x}_1)}{\|\boldsymbol{w}\|^2}\boldsymbol{w}$$

- Because $\boldsymbol{x}_1 \in \mathcal{H}$, we have $\boldsymbol{w}^T\boldsymbol{x}_1 = -b$, so $\boldsymbol{w}^T(\boldsymbol{x}_0 - \boldsymbol{x}_1) = \boldsymbol{w}^T\boldsymbol{x}_0 + b$
- Taking the Euclidean norm gives

$$d(\boldsymbol{x}_0, \mathcal{H}) = \|\boldsymbol{x}_0 - \boldsymbol{x}^*\| = \frac{|\boldsymbol{w}^T\boldsymbol{x}_0 + b|}{\|\boldsymbol{w}\|}$$



Example – Distance to a line in $\mathbb{R}^2$

Example 1 ($\mathbb{R}^2$): Find the distance from $\boldsymbol{x}_0 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ to the line $\mathcal{H} : 3x_1 - 4x_2 + 2 = 0$ and the closest point $\boldsymbol{x}^* \in \mathcal{H}$

- Here $\boldsymbol{w} = (3, -4)^T$ and $b = 2$, so $\|\boldsymbol{w}\| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$
- Signed numerator: $\boldsymbol{w}^T \boldsymbol{x}_0 + b = 3(4) - 4(3) + 2 = 2 > 0$ ($\boldsymbol{x}_0$ lies in $\mathcal{H}_+$)
- Distance: $d(\boldsymbol{x}_0, \mathcal{H}) = \frac{|2|}{5} = \frac{2}{5} = 0.4$, and closest point:

$$\boldsymbol{x}^* = \boldsymbol{x}_0 - \frac{\boldsymbol{w}^T \boldsymbol{x}_0 + b}{\|\boldsymbol{w}\|^2} \boldsymbol{w} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} - \frac{2}{25} \begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 94/25 \\ 83/25 \end{pmatrix}$$

Example – Distance to a plane in $\mathbb{R}^3$

Example 2 ($\mathbb{R}^3$): Find the distance from $\boldsymbol{x}_0 = (1, -2, 3)^T$ to the plane $\mathcal{H} : x_1 + 2x_2 + 2x_3 - 5 = 0$

- Here $\boldsymbol{w} = (1, 2, 2)^T$, $b = -5$ and $\|\boldsymbol{w}\| = \sqrt{1^2 + 2^2 + 2^2} = 3$, so

$$d(\boldsymbol{x}_0, \mathcal{H}) = \frac{|1(1) + 2(-2) + 2(3) - 5|}{3} = \frac{|-2|}{3} = \frac{2}{3}$$

Geometry of hyperplanes and support vector machines

- Hyperplanes and normal vectors
- Orthogonal distance to a hyperplane
- **Linear separability and hard-margin SVM**

Linearly separable point sets

Definition 91 (Linear separability)

Two sets of points $X_+ = \{\boldsymbol{x}_i : y_i = +1\}$ *and* $X_- = \{\boldsymbol{x}_i : y_i = -1\}$ *in* $\mathbb{R}^d$ *are* ***linearly separable*** *if there exist* $\boldsymbol{w} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$ *and* $b \in \mathbb{R}$ *such that*

$$\boldsymbol{w}^T \boldsymbol{x}_i + b > 0 \quad \text{for all } \boldsymbol{x}_i \in X_+ \qquad \text{and} \qquad \boldsymbol{w}^T \boldsymbol{x}_i + b < 0 \quad \text{for all } \boldsymbol{x}_i \in X_-$$

- Since $y_i \in \{-1, +1\}$, both inequalities can be written as a single condition:

$$y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b) > 0 \quad \text{for all } i = 1, \ldots, n$$

- Whenever a finite dataset is linearly separable, there are infinitely many separating hyperplanes – how should we pick the most robust one?

When are point sets not linearly separable? The XOR example

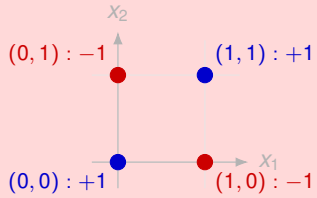
Not every two-class dataset in $\mathbb{R}^2$ is linearly separable

Example (XOR configuration):

- Let $X_+ = \{(0,0)^T, (1,1)^T\}$ $(y = +1)$ and $X_- = \{(1,0)^T, (0,1)^T\}$ $(y = -1)$
- Suppose $\boldsymbol{w}^T\boldsymbol{x} + b > 0$ on X_+ and $\boldsymbol{w}^T\boldsymbol{x} + b < 0$ on X_-:

$$\begin{aligned} \text{on } X_+ : &\quad b > 0 \text{ and } w_1 + w_2 + b > 0 \\ \text{on } X_- : &\quad w_1 + b < 0 \text{ and } w_2 + b < 0 \end{aligned}$$

- Adding the first pair gives $w_1 + w_2 + 2b > 0$, whereas adding the second pair gives $w_1 + w_2 + 2b < 0$ – contradiction!



Support vectors and canonical margin hyperplanes

Suppose $(\boldsymbol{w}, b)$ separates a finite dataset, so $\min_{1 \le i \le n} y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b) = m > 0$

Dividing both $\boldsymbol{w}$ and b by m does not change the decision boundary $\mathcal{H} = \{\boldsymbol{x} : \boldsymbol{w}^T \boldsymbol{x} + b = 0\}$, and scales the minimum functional margin to 1:

$$y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b) \ge 1 \quad \text{for all } i = 1, \ldots, n$$

In this **canonical form**, the slab free of training points is bounded by two parallel hyperplanes:

- $\mathcal{H}_{+1} = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T \boldsymbol{x} + b = +1\}$ (positive margin boundary)
- $\mathcal{H}_{-1} = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^T \boldsymbol{x} + b = -1\}$ (negative margin boundary)

The training points $\boldsymbol{x}_i$ that lie directly on $\mathcal{H}_{+1}$ or $\mathcal{H}_{-1}$ (satisfying $y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b) = 1$) are called **support vectors**

Geometric derivation of the SVM margin width

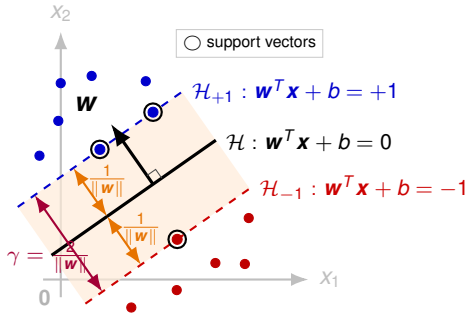
Using Theorem 90, how wide is the margin between $\mathcal{H}_{+1}$ and $\mathcal{H}_{-1}$?

- For any point $\boldsymbol{x}_+ \in \mathcal{H}_{+1}$, we have $\boldsymbol{w}^T\boldsymbol{x}_+ + b = +1$, so its distance to $\mathcal{H}$ is

$$d(\boldsymbol{x}_+, \mathcal{H}) = \frac{|\boldsymbol{w}^T\boldsymbol{x}_+ + b|}{\|\boldsymbol{w}\|} = \frac{1}{\|\boldsymbol{w}\|}$$

- For any point $\boldsymbol{x}_- \in \mathcal{H}_{-1}$, we have $\boldsymbol{w}^T\boldsymbol{x}_- + b = -1$, so its distance to $\mathcal{H}$ is also $\frac{1}{\|\boldsymbol{w}\|}$
- Therefore, the total **margin width** between $\mathcal{H}_{+1}$ and $\mathcal{H}_{-1}$ is

$$\gamma = \frac{2}{\|\boldsymbol{w}\|}$$



Hard-margin support vector machine (SVM)

To make the separating slab as wide as possible, we want to maximise $\gamma = \frac{2}{\|\boldsymbol{w}\|}$, which is equivalent to minimising $\|\boldsymbol{w}\|$ (or $\frac{1}{2}\|\boldsymbol{w}\|^2$):

$$\min_{\boldsymbol{w}\in\mathbb{R}^d,\, b\in\mathbb{R}} \frac{1}{2}\|\boldsymbol{w}\|^2 \quad \text{subject to} \quad y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1 \quad \text{for } i = 1, \ldots, n$$

- This is the **hard-margin support vector machine (SVM)** optimisation problem
- The optimal hyperplane is completely determined by the support vectors lying on the margin boundaries $\boldsymbol{w}^T\boldsymbol{x}_i + b = \pm 1$
- For small problems in $\mathbb{R}^2$, we can find $(\boldsymbol{w}, b)$ directly by **geometry and linear equations**, without any Lagrange multiplier machinery

Example 1 – Hard-margin SVM for 2 points in $\mathbb{R}^2$

Problem: Find the canonical maximum-margin hyperplane $(\boldsymbol{w}, b)$, decision boundary and margin width γ for

$$\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (y_1 = -1) \qquad \text{and} \qquad \boldsymbol{x}_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \quad (y_2 = +1)$$

Step 1 (Geometry):

- With one point in each class, both $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are support vectors, and the decision boundary $\mathcal{H}$ is the perpendicular bisector of the segment $\boldsymbol{x}_1\boldsymbol{x}_2$
- The normal vector $\boldsymbol{w}$ must point from $\boldsymbol{x}_1$ ($y_1 = -1$) towards $\boldsymbol{x}_2$ ($y_2 = +1$), so $\boldsymbol{w}$ is a positive multiple of $\boldsymbol{x}_2 - \boldsymbol{x}_1$:

$$\boldsymbol{x}_2 - \boldsymbol{x}_1 = \begin{pmatrix} 3-1 \\ 5-1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad \Longrightarrow \quad \boldsymbol{w} = \alpha \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (\alpha > 0)$$

Example 1 – Scaling $(\boldsymbol{w}, b)$ and computing the margin

Step 2 (Canonical equations): Since $\boldsymbol{x}_1 \in \mathcal{H}_{-1}$ and $\boldsymbol{x}_2 \in \mathcal{H}_{+1}$, we require

$$\boldsymbol{w}^T \boldsymbol{x}_1 + b = -1 \qquad \text{and} \qquad \boldsymbol{w}^T \boldsymbol{x}_2 + b = +1$$

- Subtracting the first equation from the second gives $\boldsymbol{w}^T(\boldsymbol{x}_2 - \boldsymbol{x}_1) = 2$:

$$\alpha \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \iff 10\alpha = 2 \iff \alpha = \frac{1}{5} \quad \Longrightarrow \quad \boldsymbol{w} = \begin{pmatrix} 1/5 \\ 2/5 \end{pmatrix}$$

- Substituting $\boldsymbol{w}$ into $\boldsymbol{w}^T \boldsymbol{x}_1 + b = -1$ gives the offset b:

$$\frac{1}{5}(1) + \frac{2}{5}(1) + b = -1 \quad \Longrightarrow \quad b = -1 - \frac{3}{5} = -\frac{8}{5}$$

Step 3 (Decision boundary and margin width):

- Decision boundary: $\frac{1}{5}x_1 + \frac{2}{5}x_2 - \frac{8}{5} = 0 \iff x_1 + 2x_2 - 8 = 0$ (passes through the midpoint $\boldsymbol{m} = (2, 3)^T$)
- Margin width: $\|\boldsymbol{w}\| = \sqrt{(1/5)^2 + (2/5)^2} = \frac{1}{\sqrt{5}}$, so $\gamma = \frac{2}{\|\boldsymbol{w}\|} = 2\sqrt{5} = \|\boldsymbol{x}_2 - \boldsymbol{x}_1\|$

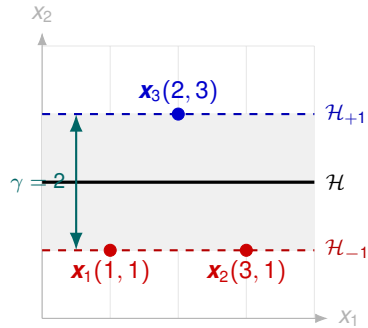
Example 2 – Hard-margin SVM for 3 points in $\mathbb{R}^2$

Problem: Find $(\boldsymbol{w}, b)$ and margin γ for:

- Class -1: $\boldsymbol{x}_1 = (1,1)^T$ and $\boldsymbol{x}_2 = (3,1)^T$
- Class $+1$: $\boldsymbol{x}_3 = (2,3)^T$

Step 1 (Direction of the boundary):

- Both $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ belong to class -1 and are equidistant from $\boldsymbol{x}_3$, so all three points are support vectors
- Since $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{H}_{-1}$, the vector $\boldsymbol{x}_2 - \boldsymbol{x}_1 = (2,0)^T$ lies along $\mathcal{H}_{-1}$
- By orthogonality, $\boldsymbol{w}^T(\boldsymbol{x}_2 - \boldsymbol{x}_1) = 2w_1 = 0 \implies w_1 = 0$, so $\boldsymbol{w} = (0, w_2)^T$



Example 2 – Solving the linear equations

Step 2 (Canonical equations): Substituting $\boldsymbol{w} = (0, w_2)^T$ into $\boldsymbol{w}^T \boldsymbol{x}_i + b = y_i$ for the support vectors gives

$$\begin{aligned} \text{at } \boldsymbol{x}_1 = (1,1)^T \text{ and } \boldsymbol{x}_2 = (3,1)^T: \quad & w_2(1) + b = -1 \\ \text{at } \boldsymbol{x}_3 = (2,3)^T: \quad & w_2(3) + b = +1 \end{aligned}$$

- Subtracting the first equation from the second gives $2w_2 = 2 \implies w_2 = 1$
- Substituting $w_2 = 1$ into $w_2 + b = -1$ gives $b = -2$, so

$$\boldsymbol{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \text{and} \qquad b = -2$$

- Decision boundary: $x_2 - 2 = 0$ (the horizontal line $x_2 = 2$)
- Margin width: $\|\boldsymbol{w}\| = 1 \implies \gamma = \frac{2}{\|\boldsymbol{w}\|} = 2$

What if we add non-support-vector points?

Suppose we add two more points to Example 2:

$$\boldsymbol{x}_4 = \begin{pmatrix} 2 \\ 5 \end{pmatrix} \quad (y_4 = +1) \qquad \text{and} \qquad \boldsymbol{x}_5 = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad (y_5 = -1)$$

- Check the functional margin $y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b) = y_i(x_{i2} - 2)$ at the new points:

$$y_4(\boldsymbol{w}^T \boldsymbol{x}_4 + b) = (+1)(5 - 2) = 3 > 1$$
$$y_5(\boldsymbol{w}^T \boldsymbol{x}_5 + b) = (-1)(-1 - 2) = 3 > 1$$

- Because both points lie strictly outside the margin slab (with $y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b) > 1$), they are **not support vectors**
- Moving or deleting $\boldsymbol{x}_4$ and $\boldsymbol{x}_5$ (as long as they stay outside the margin) does **not** change $(\boldsymbol{w}, b)$ at all!

Supervised vs unsupervised linear methods

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Framing classification as linear least squares

Can we fit a linear classifier $\hat{y}(\boldsymbol{x}) = \operatorname{sgn}(\boldsymbol{w}^T\boldsymbol{x} + b)$ using the method of least squares from Unit 03?

We ask the affine score $f(\boldsymbol{x}_i) = \boldsymbol{w}^T\boldsymbol{x}_i + b$ to match the numeric class label $y_i \in \{-1, +1\}$ as closely as possible:

$$\boldsymbol{w}^T\boldsymbol{x}_i + b \approx y_i \quad \text{for } i = 1, \ldots, n$$

To write this as a single matrix product, absorb the intercept b by defining the **augmented feature vector** and **augmented weight vector**:

$$\tilde{\boldsymbol{x}}_i = \begin{pmatrix} \boldsymbol{x}_i \\ 1 \end{pmatrix} \in \mathbb{R}^{d+1}, \qquad \tilde{\boldsymbol{w}} = \begin{pmatrix} \boldsymbol{w} \\ b \end{pmatrix} \in \mathbb{R}^{d+1}$$

so that $\boldsymbol{w}^T\boldsymbol{x}_i + b = \tilde{\boldsymbol{x}}_i^T\tilde{\boldsymbol{w}}$

Matrix formulation and normal equations

Stack the augmented rows $\tilde{\boldsymbol{x}}_i^T$ into the design matrix $X \in \mathcal{M}_{n,d+1}$ and the labels into $\boldsymbol{y} \in \{-1, +1\}^n$:

$$X = \begin{pmatrix} \boldsymbol{x}_1^T & 1 \\ \vdots & \vdots \\ \boldsymbol{x}_n^T & 1 \end{pmatrix} \in \mathcal{M}_{n,d+1}, \qquad \boldsymbol{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n$$

Minimising the sum of squared errors $\|X\tilde{\boldsymbol{w}} - \boldsymbol{y}\|^2 = \sum_{i=1}^n (\boldsymbol{w}^T \boldsymbol{x}_i + b - y_i)^2$ yields the **normal equations**:

$$X^T X \tilde{\boldsymbol{w}} = X^T \boldsymbol{y} \quad \Longrightarrow \quad \tilde{\boldsymbol{w}} = (X^T X)^{-1} X^T \boldsymbol{y}$$

- What we are explaining here **only works when we have more observations than features**: we require $n \geq d + 1$ and $\text{rank}(X) = d + 1$ for $X^T X \in \mathcal{M}_{d+1}$ to be invertible
- If $n < d + 1$ or if features are collinear, $X^T X$ is singular and $(X^T X)^{-1}$ does not exist

Example – 4-point least-squares classifier in $\mathbb{R}^2$

Problem: Find the least-squares linear classifier $(\boldsymbol{w}, b)$ for the $n = 4$ points:

- Class -1 ($y_1 = y_2 = -1$): $\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ and $\boldsymbol{x}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$
- Class $+1$ ($y_3 = y_4 = +1$): $\boldsymbol{x}_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\boldsymbol{x}_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Step 1 (Set up X and $\boldsymbol{y}$):

$$X = \begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in \mathcal{M}_{4,3}, \qquad \boldsymbol{y} = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

Example – Solving the normal equations by hand

Step 2 (Compute $X^T X$ and $X^T \boldsymbol{y}$):

$$X^T X = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad X^T \boldsymbol{y} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$$

Step 3 (Solve for $\tilde{\boldsymbol{w}}$): Since $X^T X = \mathrm{diag}(2, 2, 4)$ is diagonal, its inverse is immediate:

$$\tilde{\boldsymbol{w}} = (X^T X)^{-1} X^T \boldsymbol{y} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/4 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \implies \boldsymbol{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b = 0$$

The decision boundary is $\boldsymbol{w}^T \boldsymbol{x} + b = x_1 + x_2 = 0$ (the line $x_2 = -x_1$), which separates the two classes cleanly

Example – Fitted values, residuals and classifying new points

With $\tilde{\boldsymbol{w}} = (1, 1, 0)^T$ ($\boldsymbol{w} = (1, 1)^T$ and $b = 0$), let us check the fitted scores $\hat{\boldsymbol{y}} = X\tilde{\boldsymbol{w}}$ and residual vector $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}}$:

$$\hat{\boldsymbol{y}} = X\tilde{\boldsymbol{w}} = \begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 1 \end{pmatrix} = \boldsymbol{y} \quad \Longrightarrow \quad \boldsymbol{e} = \boldsymbol{y} - X\tilde{\boldsymbol{w}} = \boldsymbol{0}$$

- Because these 4 points happen to lie on the lines $x_1 + x_2 = -1$ and $x_1 + x_2 = +1$, the least-squares residual is zero!
- For a new observation $\boldsymbol{x}_{\text{new}} = \begin{pmatrix} 2 \\ -0.5 \end{pmatrix}$, we compute $\boldsymbol{w}^T\boldsymbol{x}_{\text{new}} + b = 2 - 0.5 = 1.5 > 0$, so $\hat{y}(\boldsymbol{x}_{\text{new}}) = \text{sgn}(1.5) = +1$

Sensitivity of least-squares classification to outliers

Why is least squares not always suitable for classification?

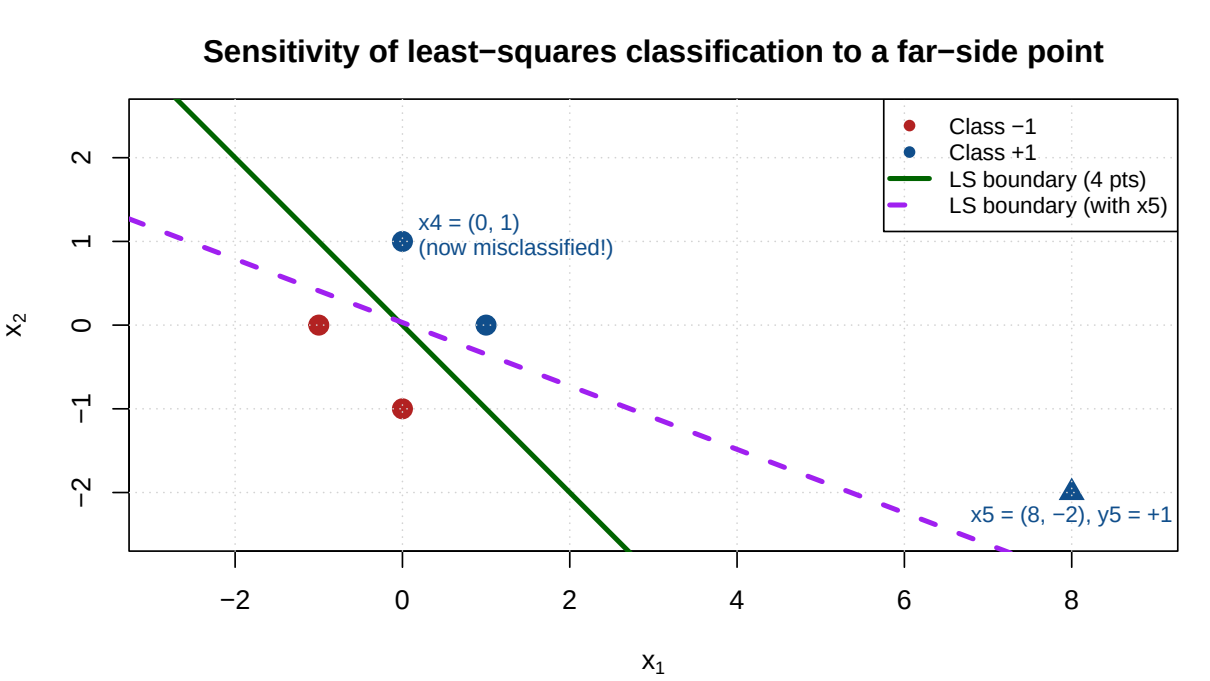
- The squared error $(\boldsymbol{w}^T \boldsymbol{x}_i + b - y_i)^2$ penalises predictions that are **too far on the correct side** of the decision boundary
- Suppose a point with $y_i = +1$ lies deep inside $\mathcal{H}_+$ so that $\boldsymbol{w}^T \boldsymbol{x}_i + b = 10 > 0$: it is *correctly classified*, yet least squares assigns a large squared penalty $(10 - 1)^2 = 81$
- To shrink that large residual, least squares tilts and shifts the hyperplane towards the extreme point, which can push the boundary across points near the margin and cause misclassifications

R: least-squares classifier and outlier sensitivity

Compare the 4-point least-squares boundary before and after adding a single "easy" Class $+1$ point at $\boldsymbol{x}_5 = (8, -2)^T$ (which lies well inside $x_1 + x_2 > 0$)

```
X4 = matrix(c(-1, 0, 1,  0, -1, 1,  1, 0, 1,  0, 1, 1),
            nrow = 4, byrow = TRUE)
y4 = c(-1, -1, 1, 1)
w_tilde_4 = solve(t(X4) %*% X4, t(X4) %*% y4)
# Add one outlier far on the correct (+1) side: x5 = (8, -2)^T
X5 = rbind(X4, c(8, -2, 1)); y5 = c(y4, 1)
w_tilde_5 = solve(t(X5) %*% X5, t(X5) %*% y5)
cbind(w_4pts = round(w_tilde_4, 3), w_5pts = round(w_tilde_5, 3))
```

```
##      [,1]   [,2]
## [1,]    1  0.433
## [2,]    1  1.142
## [3,]    0 -0.035
```


Sensitivity of least−squares classification to a far−side point
Class −1
Class +1
LS boundary (4 pts)
LS boundary (with x5)
x4 = (0, 1)
(now misclassified!)
x5 = (8, −2), y5 = +1
x1
x2
−2
0
2
4
6
8
−2
−1
0
1
2

Ridge regularisation when $X^T X$ is singular

What if we have more features than observations ($n < d + 1$) or collinear features?

- In that case, $\text{rank}(X) < d + 1$, so $X^T X$ is singular and cannot be inverted
- To fix this, we add a **ridge penalty** with regularisation parameter $\lambda > 0$:

$$\min_{\tilde{\boldsymbol{w}} \in \mathbb{R}^{d+1}} \left(\|X\tilde{\boldsymbol{w}} - \boldsymbol{y}\|^2 + \lambda \|\tilde{\boldsymbol{w}}\|^2 \right)$$

- Setting the gradient to zero yields the regularised normal equations:

$$(X^T X + \lambda \mathbb{I})\tilde{\boldsymbol{w}} = X^T \boldsymbol{y} \quad \Longrightarrow \quad \tilde{\boldsymbol{w}}_{\text{ridge}} = (X^T X + \lambda \mathbb{I})^{-1} X^T \boldsymbol{y}$$

- Adding $\lambda \mathbb{I}$ shifts all eigenvalues of $X^T X$ by $+\lambda > 0$: all its eigenvalues are strictly positive (we call such a matrix **positive definite**; we will discuss this in more detail later), which guarantees that it is always invertible

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Fisher's linear discriminant analysis (LDA)

Projection onto a line and class separation

Scatter matrices and the Rayleigh quotient

Example and R implementation

Fisher's idea (1936) – Supervised 1D projection

Instead of forcing linear outputs to equal ± 1, Ronald A. Fisher (1936) framed binary classification as **supervised dimensionality reduction**:

- Suppose our n training observations in $\mathbb{R}^d$ are partitioned into two classes:
  - Class C_1 with n_1 points $\{\boldsymbol{x}_i : i \in C_1\}$
  - Class C_2 with n_2 points $\{\boldsymbol{x}_i : i \in C_2\}$, where $n_1 + n_2 = n$

- Project each vector $\boldsymbol{x}_i \in \mathbb{R}^d$ onto a 1D line directed by $\boldsymbol{w} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}$ via the scalar dot product

$$z_i = \boldsymbol{w}^T \boldsymbol{x}_i \in \mathbb{R}$$

- Goal: choose the direction of $\boldsymbol{w}$ so that the projected scalars $\{z_i : i \in C_1\}$ and $\{z_i : i \in C_2\}$ are separated as cleanly as possible

Class sample means before and after projection

The sample mean vectors (centroids) of the two classes in $\mathbb{R}^d$ are

$$\boldsymbol{\mu}_1 = \frac{1}{n_1} \sum_{i \in C_1} \boldsymbol{x}_i \in \mathbb{R}^d \qquad \text{and} \qquad \boldsymbol{\mu}_2 = \frac{1}{n_2} \sum_{i \in C_2} \boldsymbol{x}_i \in \mathbb{R}^d$$

By linearity of matrix multiplication, the sample means of the 1D projected points $z_i = \boldsymbol{w}^T \boldsymbol{x}_i$ are simply the projections of $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$:

$$\tilde{\mu}_1 = \frac{1}{n_1} \sum_{i \in C_1} z_i = \frac{1}{n_1} \sum_{i \in C_1} \boldsymbol{w}^T \boldsymbol{x}_i = \boldsymbol{w}^T \boldsymbol{\mu}_1 \qquad \text{and} \qquad \tilde{\mu}_2 = \boldsymbol{w}^T \boldsymbol{\mu}_2$$

Hence the squared distance between the projected class means is

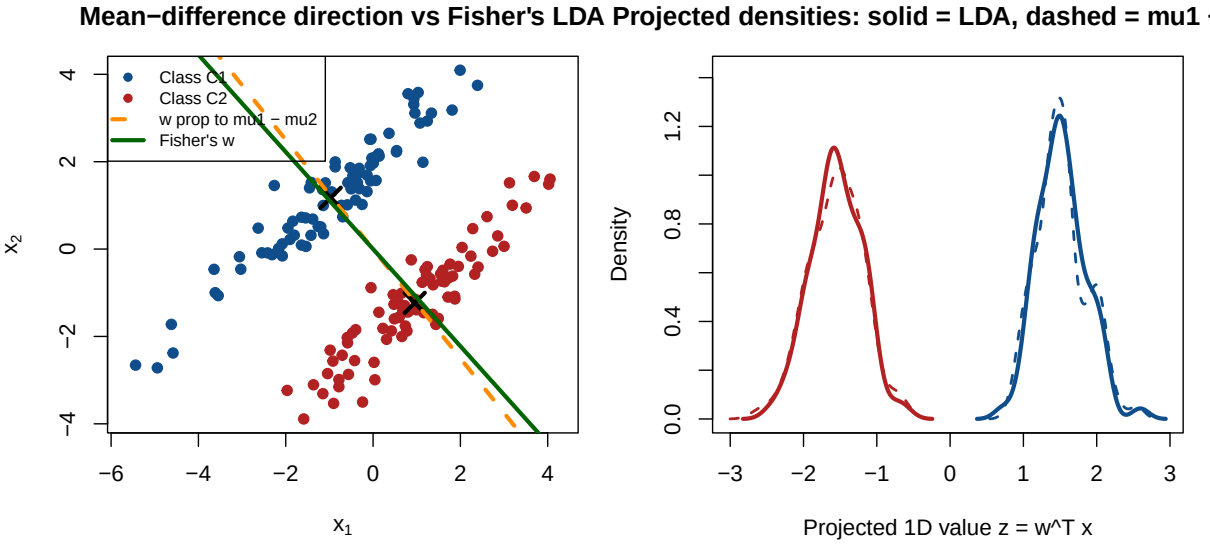
$$(\tilde{\mu}_1 - \tilde{\mu}_2)^2 = \Big(\boldsymbol{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \Big)^2$$

Why maximising mean separation alone is not enough

A naive first idea is to choose a unit vector $\boldsymbol{w}$ (*i.e.*, $\|\boldsymbol{w}\| = 1$) that maximises the squared distance between projected means:

$$(\tilde{\mu}_1 - \tilde{\mu}_2)^2 = \left(\boldsymbol{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\right)^2$$

- By the Cauchy–Schwarz inequality, $|\boldsymbol{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)| \leq \|\boldsymbol{w}\| \, \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|$, with equality when $\boldsymbol{w} \propto \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ (the line joining the two centroids)
- **Problem:** $\boldsymbol{w} \propto \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ completely ignores the covariance (shape and orientation) of the point clouds within each class!
- If the classes are tilted ellipses, projecting onto $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ can produce large within-class spread and heavy overlap, whereas another direction has a smaller gap between means but much tighter within-class clusters


Mean−difference direction vs Fisher's LDA Projected densities: solid = LDA, dashed = mu1
Class C1
Class C2
w prop to mu1 − mu2
Fisher's w
x2
x1
−6
−4
−2
0
2
4
Density
0.0
0.4
0.8
1.2
−3
−2
−1
0
1
2
3
Projected 1D value z = w^T x

Fisher's linear discriminant analysis (LDA)

- Projection onto a line and class separation
- **Scatter matrices and the Rayleigh quotient**
- Example and R implementation

Within-class scatter of the projected data

Fisher's idea: maximise the squared distance between projected class means **relative to** the within-class spread of the projected points

For each class C_k ($k \in \{1, 2\}$), define the **projected within-class scatter** as the sum of squared deviations around the projected mean $\tilde{\mu}_k$:

$$\tilde{s}_k^2 = \sum_{i \in C_k} (z_i - \tilde{\mu}_k)^2 = \sum_{i \in C_k} \left(\boldsymbol{w}^T \boldsymbol{x}_i - \boldsymbol{w}^T \boldsymbol{\mu}_k \right)^2 = \sum_{i \in C_k} \left(\boldsymbol{w}^T (\boldsymbol{x}_i - \boldsymbol{\mu}_k) \right)^2$$

(Notice that $\frac{1}{n_k - 1} \tilde{s}_k^2$ is the sample variance of z_i within class C_k)

Fisher's criterion chooses $\boldsymbol{w} \neq \mathbf{0}$ to maximise the signal-to-noise ratio

$$J(\boldsymbol{w}) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

Within-class and between-class scatter matrices

Definition 92 (Scatter matrices)

Let C_1 and C_2 be two classes in $\mathbb{R}^d$ with sample means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$

1. *The* ***within-class scatter matrix*** *for class C_k ($k \in \{1, 2\}$) is*

$$S_k = \sum_{i \in C_k} (\boldsymbol{x}_i - \boldsymbol{\mu}_k)(\boldsymbol{x}_i - \boldsymbol{\mu}_k)^T \in \mathcal{M}_d$$

2. *The* ***total within-class scatter matrix*** *is $S_W = S_1 + S_2 \in \mathcal{M}_d$*
3. *The* ***between-class scatter matrix*** *is*

$$S_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \in \mathcal{M}_d$$

If $X_{c,k} \in \mathcal{M}_{n_k,d}$ has rows $(\boldsymbol{x}_i - \boldsymbol{\mu}_k)^T$ for $i \in C_k$, then $S_k = X_{c,k}^T X_{c,k} = (n_k - 1)\Sigma_k$

Writing Fisher's criterion as a generalised Rayleigh quotient

Using $(\boldsymbol{w}^T\boldsymbol{u})^2 = (\boldsymbol{w}^T\boldsymbol{u})(\boldsymbol{u}^T\boldsymbol{w}) = \boldsymbol{w}^T(\boldsymbol{u}\boldsymbol{u}^T)\boldsymbol{w}$ for any vector $\boldsymbol{u} \in \mathbb{R}^d$:

- The numerator of $J(\boldsymbol{w})$ becomes a quadratic form with matrix S_B:

$$(\tilde{\mu}_1 - \tilde{\mu}_2)^2 = \Big(\boldsymbol{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\Big)^2 = \boldsymbol{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T\boldsymbol{w} = \boldsymbol{w}^T S_B \boldsymbol{w}$$

- For each class C_k, the projected scatter is $\tilde{s}_k^2 = \boldsymbol{w}^T S_k \boldsymbol{w}$, so the denominator becomes a quadratic form with matrix $S_W = S_1 + S_2$:

$$\tilde{s}_1^2 + \tilde{s}_2^2 = \boldsymbol{w}^T S_1 \boldsymbol{w} + \boldsymbol{w}^T S_2 \boldsymbol{w} = \boldsymbol{w}^T(S_1 + S_2)\boldsymbol{w} = \boldsymbol{w}^T S_W \boldsymbol{w}$$

Therefore, Fisher's criterion is the **generalised Rayleigh quotient**

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}$$

Linear algebraic properties of S_B and S_W

Proposition 93 (Properties of S_B and S_W)

Assume $\boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$. The scatter matrices $S_B, S_W \in \mathcal{M}_d$ satisfy:

1. ***Symmetry:*** $S_B^T = S_B$ *and* $S_W^T = S_W$
2. ***Positive semi-definiteness:*** $\boldsymbol{w}^T S_B \boldsymbol{w} \geq 0$ *and* $\boldsymbol{w}^T S_W \boldsymbol{w} \geq 0$ *for all* $\boldsymbol{w} \in \mathbb{R}^d$
3. ***Rank-1 structure of*** S_B***:*** $\operatorname{rank}(S_B) = 1$ *and* $Im\,(S_B) = \operatorname{span}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$*; specifically, for every* $\boldsymbol{w} \in \mathbb{R}^d$*,*

$$S_B \boldsymbol{w} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \underbrace{\left((\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \boldsymbol{w} \right)}_{\text{scalar}}$$

4. ***Invertibility of*** S_W***:*** *if the centred observations* $\{\boldsymbol{x}_i - \boldsymbol{\mu}_{y_i}\}$ *span* $\mathbb{R}^d$ *(which requires* $n - 2 \geq d$*), then* S_W *is positive definite and invertible*

Fisher's linear discriminant theorem

Theorem 94 (Fisher's linear discriminant)

Suppose $\boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$ and S_W is invertible. The generalised Rayleigh quotient

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}$$

attains its maximum when $\boldsymbol{w}$ satisfies the generalised eigenvalue problem $S_B\boldsymbol{w} = \lambda S_W \boldsymbol{w}$, and the optimal projection direction is given up to a non-zero scalar multiple by

$$\boldsymbol{w}^* \propto S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

- Notice that $J(\alpha\boldsymbol{w}) = \frac{\alpha^2 \boldsymbol{w}^T S_B \boldsymbol{w}}{\alpha^2 \boldsymbol{w}^T S_W \boldsymbol{w}} = J(\boldsymbol{w})$ for every $\alpha \neq 0$: only the **direction** of $\boldsymbol{w}^*$ matters, not its length

Proof – Maximising the Rayleigh quotient (Part 1)

Recall from Unit 05 that for any symmetric matrix $A \in \mathcal{M}_d$, the gradient of the quadratic form $q(\boldsymbol{w}) = \boldsymbol{w}^T A \boldsymbol{w}$ is $\nabla q(\boldsymbol{w}) = 2A\boldsymbol{w}$

- Differentiating $J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}$ with respect to $\boldsymbol{w}$ via the quotient rule and setting $\nabla J(\boldsymbol{w}) = \boldsymbol{0}$ gives

$$\nabla J(\boldsymbol{w}) = \frac{2S_B\boldsymbol{w}(\boldsymbol{w}^T S_W \boldsymbol{w}) - (\boldsymbol{w}^T S_B \boldsymbol{w})(2S_W \boldsymbol{w})}{(\boldsymbol{w}^T S_W \boldsymbol{w})^2} = \boldsymbol{0}$$

- Since $\boldsymbol{w}^T S_W \boldsymbol{w} > 0$, dividing by $2(\boldsymbol{w}^T S_W \boldsymbol{w})$ and setting $\lambda = J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}$ yields the generalised eigenvalue equation

$$S_B\boldsymbol{w} = \lambda S_W \boldsymbol{w} \quad \iff \quad S_W^{-1} S_B \boldsymbol{w} = \lambda \boldsymbol{w}$$

(Equivalently, maximising $\boldsymbol{w}^T S_B \boldsymbol{w}$ subject to $\boldsymbol{w}^T S_W \boldsymbol{w} = 1$ with Lagrange multiplier λ gives $2S_B\boldsymbol{w} = 2\lambda S_W \boldsymbol{w}$ immediately)

Proof – Solving for $\boldsymbol{w}^*$ without eigensolvers (Part 2)

Why do we not need to compute eigenvalues and eigenvectors of $S_W^{-1} S_B$?

- Because $S_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T$ has rank 1! For any vector $\boldsymbol{w} \in \mathbb{R}^d$,

$$S_B \boldsymbol{w} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \underbrace{\left((\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \boldsymbol{w} \right)}_{= c \in \mathbb{R}} = c(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

  so $S_B \boldsymbol{w}$ always points in the direction of $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$

- Substituting $S_B \boldsymbol{w} = c(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ into $S_B \boldsymbol{w} = \lambda S_W \boldsymbol{w}$ (with $\lambda > 0$) and multiplying on the left by S_W^{-1} gives

$$\lambda S_W \boldsymbol{w} = c(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \quad \Longrightarrow \quad \boldsymbol{w} = \frac{c}{\lambda} S_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

- Since $\frac{c}{\lambda}$ is a non-zero scalar and $J(\boldsymbol{w})$ is scale-invariant, we can drop the scalar factor and take $\boldsymbol{w}^* = S_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ □

Special case and classification threshold

Special case (isotropic within-class scatter):

- If $S_W = \sigma^2 \mathbb{I}$ (both classes are spherical with zero correlation between features), then

$$\boldsymbol{w}^* \propto (\sigma^2 \mathbb{I})^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \propto \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$$

  In general, S_W^{-1} rotates the direction away from $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ to account for feature correlations and unequal variances

Choosing the threshold b to classify new points:

- With $\boldsymbol{w} = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, since S_W^{-1} is positive definite, $\tilde{\mu}_1 - \tilde{\mu}_2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) > 0$, so $\tilde{\mu}_1 > \tilde{\mu}_2$
- Placing the 1D cutoff halfway between $\tilde{\mu}_1$ and $\tilde{\mu}_2$ gives the offset

$$b = -\frac{1}{2}(\tilde{\mu}_1 + \tilde{\mu}_2) = -\frac{1}{2}\boldsymbol{w}^T(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$$

  and we classify $\boldsymbol{x}_{\text{new}}$ into C_1 if $\boldsymbol{w}^T\boldsymbol{x}_{\text{new}} + b > 0$ and into C_2 if $\boldsymbol{w}^T\boldsymbol{x}_{\text{new}} + b < 0$

Fisher’s linear discriminant analysis (LDA)

- Projection onto a line and class separation
- Scatter matrices and the Rayleigh quotient
- **Example and R implementation**

Example – Fisher's LDA by hand in $\mathbb{R}^2$

Problem: Consider $n = 6$ points in $\mathbb{R}^2$ split into two classes of $n_1 = n_2 = 3$ points:

$$\text{Class } C_1: \quad \boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \boldsymbol{x}_2 = \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \quad \boldsymbol{x}_3 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\text{Class } C_2: \quad \boldsymbol{x}_4 = \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad \boldsymbol{x}_5 = \begin{pmatrix} 6 \\ 3 \end{pmatrix}, \quad \boldsymbol{x}_6 = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$

1. Compute the class means $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ and scatter matrices S_1, S_2, S_W
2. Find Fisher's optimal direction $\boldsymbol{w} = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ and compare $J(\boldsymbol{w})$ with $J(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$
3. Find the midpoint threshold $b = -\frac{1}{2}\boldsymbol{w}^T(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$ and classify $\boldsymbol{x}_{\text{new}} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$

Step 1 – Class sample means and centred deviations

Compute the class sample means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$:

$$\boldsymbol{\mu}_1 = \frac{1}{3}\begin{pmatrix} 1+2+3 \\ 3+5+4 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \qquad \boldsymbol{\mu}_2 = \frac{1}{3}\begin{pmatrix} 5+6+7 \\ 1+3+2 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$

so the mean difference vector is $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$

Subtract each class mean from the points in that class:

- For C_1: $\boldsymbol{x}_1 - \boldsymbol{\mu}_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, $\boldsymbol{x}_2 - \boldsymbol{\mu}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\boldsymbol{x}_3 - \boldsymbol{\mu}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- For C_2: $\boldsymbol{x}_4 - \boldsymbol{\mu}_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, $\boldsymbol{x}_5 - \boldsymbol{\mu}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\boldsymbol{x}_6 - \boldsymbol{\mu}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Step 2 – Within-class scatter matrices S_1, S_2 and S_W

Compute $S_1 = \sum_{i \in C_1} (\boldsymbol{x}_i - \boldsymbol{\mu}_1)(\boldsymbol{x}_i - \boldsymbol{\mu}_1)^T$:

$$S_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \begin{pmatrix} -1 & -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}$$
$$= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Since C_2 has the identical deviation vectors around $\boldsymbol{\mu}_2$, we also have $S_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, so the total within-class scatter matrix is

$$S_W = S_1 + S_2 = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$$

Step 3 – Inverting S_W and finding Fisher's direction $\boldsymbol{w}$

For the 2×2 matrix $S_W = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$, we have $\det(S_W) = 4(4) - 2(2) = 12 \neq 0$, so

$$S_W^{-1} = \frac{1}{12} \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

By Theorem 94, Fisher's optimal discriminant vector is

$$\boldsymbol{w}^* = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) = \frac{1}{6} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} -4 \\ 2 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} -10 \\ 8 \end{pmatrix} = \begin{pmatrix} -5/3 \\ 4/3 \end{pmatrix}$$

Rescaling by 3 to clear fractions gives the integer direction vector

$$\boldsymbol{w} = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$$

Compare the slopes: $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = (-4, 2)^T$ has slope $-1/2 = -0.5$, whereas $\boldsymbol{w} = (-5, 4)^T$ has slope $-4/5 = -0.8$

Step 4 – Verifying the 1D projected values directly

Let us compute the 1D projections $z_i = \boldsymbol{w}^T \boldsymbol{x}_i = -5x_{i1} + 4x_{i2}$ of all 6 points using $\boldsymbol{w} = (-5, 4)^T$:

- **Class C_1:**

$$z_1 = -5(1) + 4(3) = 7, \;\; z_2 = -5(2) + 4(5) = 10, \;\; z_3 = -5(3) + 4(4) = 1$$

Projected mean and scatter: $\tilde{\mu}_1 = \frac{7+10+1}{3} = 6 = \boldsymbol{w}^T \boldsymbol{\mu}_1$ and
$\tilde{s}_1^2 = (7-6)^2 + (10-6)^2 + (1-6)^2 = 1 + 16 + 25 = 42$

- **Class C_2:**

$$z_4 = -5(5)+4(1) = -21, \; z_5 = -5(6)+4(3) = -18, \; z_6 = -5(7)+4(2) = -27$$

Projected mean and scatter: $\tilde{\mu}_2 = \frac{-21-18-27}{3} = -22 = \boldsymbol{w}^T \boldsymbol{\mu}_2$ and
$\tilde{s}_2^2 = 1^2 + 4^2 + (-5)^2 = 1 + 16 + 25 = 42$

Total 1D scatter: $\tilde{s}_1^2 + \tilde{s}_2^2 = 42 + 42 = 84 = \boldsymbol{w}^T S_W \boldsymbol{w}$

Step 5 – Comparing Rayleigh quotients and classifying $\boldsymbol{x}_{\text{new}}$

Comparing $J(\boldsymbol{w})$ vs $J(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$:

- For $\boldsymbol{w} = (-5, 4)^T$: $(\tilde{\mu}_1 - \tilde{\mu}_2)^2 = (6 - (-22))^2 = 28^2 = 784$ and $\boldsymbol{w}^T S_W \boldsymbol{w} = 84$, so
$$J(\boldsymbol{w}) = \frac{784}{84} = \frac{28}{3} \approx 9.333$$
- For $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = (-4, 2)^T$: $J(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) = \frac{20^2}{48} = \frac{400}{48} = \frac{25}{3} \approx 8.333 < \frac{28}{3}$

Threshold b and classifying $\boldsymbol{x}_{\text{new}} = (4, 4)^T$ (using $\boldsymbol{w} = (-5, 4)^T$):

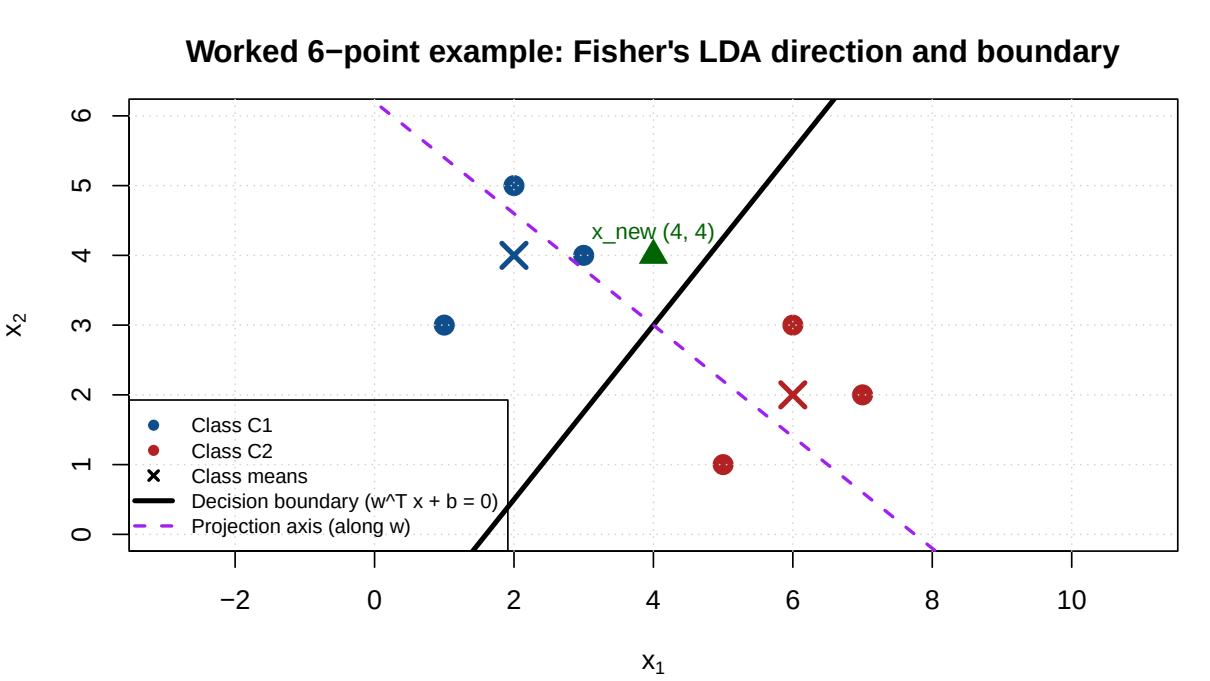
- Midpoint of means: $\frac{1}{2}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$, so $b = -\boldsymbol{w}^T \begin{pmatrix} 4 \\ 3 \end{pmatrix} = -(-20 + 12) = 8$
- Decision boundary: $-5x_1 + 4x_2 + 8 = 0$
- At $\boldsymbol{x}_{\text{new}} = (4, 4)^T$: $\boldsymbol{w}^T \boldsymbol{x}_{\text{new}} + b = -5(4) + 4(4) + 8 = +4 > 0 \implies$ **Class C_1**

R: verifying the 6-point example

In R, since $\text{cov}(X_k) = \frac{1}{n_k - 1} S_k$, we have $S_k = (n_k - 1)\,\texttt{cov}(X_k)$:

```
X1 = matrix(c(1, 3,  2, 5,  3, 4), nrow = 3, byrow = TRUE)
X2 = matrix(c(5, 1,  6, 3,  7, 2), nrow = 3, byrow = TRUE)
mu1 = colMeans(X1); mu2 = colMeans(X2)
SW = (nrow(X1) - 1) * cov(X1) + (nrow(X2) - 1) * cov(X2)
w_star = solve(SW, mu1 - mu2)
w = 3 * w_star
b = -0.5 * sum(w * (mu1 + mu2))
c(w1 = w[1], w2 = w[2], b = b, score_new = sum(w * c(4, 4)) + b)

##        w1        w2         b score_new
##        -5         4         8         4
```

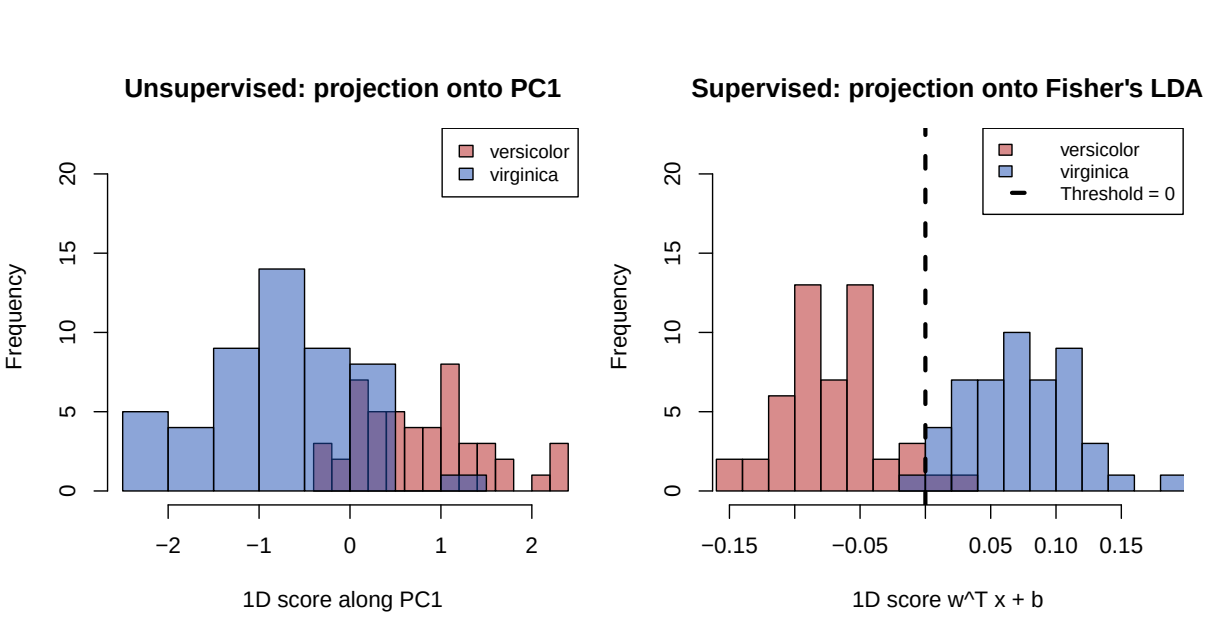

Worked 6−point example: Fisher's LDA direction and boundary
x_new (4, 4)
Class C1
Class C2
Class means
Decision boundary (w^T x + b = 0)
Projection axis (along w)
x2
x1
0
1
2
3
4
5
6
−2
0
2
4
6
8
10

PCA vs Fisher's LDA on the Iris dataset

Compare unsupervised PCA and supervised Fisher's LDA for separating *Iris versicolor* and *Iris virginica* using all $d = 4$ features:

```
iris_sub = subset(iris, Species %in% c("versicolor", "virginica"))
X_iris = as.matrix(iris_sub[, 1:4])
y_iris = ifelse(iris_sub$Species == "virginica", 1, -1)
pc1_iris = eigen(cov(X_iris))$vectors[, 1] # Unsupervised PC1
X_c1 = X_iris[y_iris == 1, ];  X_c2 = X_iris[y_iris == -1, ]
mu_1 = colMeans(X_c1);         mu_2 = colMeans(X_c2)
SW_iris = (nrow(X_c1) - 1) * cov(X_c1) + (nrow(X_c2) - 1) * cov(X_c2)
w_iris = solve(SW_iris, mu_1 - mu_2)       # Supervised Fisher LDA
b_iris = -0.5 * sum(w_iris * (mu_1 + mu_2))
mean(sign(as.vector(X_iris %*% w_iris) + b_iris) == y_iris)

## [1] 0.97
```


Unsupervised: projection onto PC1
versicolor
virginica
Frequency
0
5
10
15
20
−2
−1
0
1
2
1D score along PC1
Supervised: projection onto Fisher's LDA
versicolor
virginica
Threshold = 0
Frequency
0
5
10
15
20
−0.15
−0.05
0.05
0.10
0.15
1D score w^T x + b

Summary – Comparing linear projection and classification methods

Method	Supervised?	Criterion optimised	Linear algebraic solution
PCA (Unit 05)	No	Max variance $\boldsymbol{w}^T \Sigma \boldsymbol{w}$	Top eigenvector of Σ
Hard-margin SVM	Yes	Max margin $\frac{2}{\|\boldsymbol{w}\|}$	Support vectors $\boldsymbol{w}^T \boldsymbol{x}_i + b = \pm 1$
Least squares	Yes	Min error $\|X\tilde{\boldsymbol{w}} - \boldsymbol{y}\|^2$	$\tilde{\boldsymbol{w}} = (X^T X)^{-1} X^T \boldsymbol{y}$
Fisher's LDA	Yes	Max quotient $\frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}$	$\boldsymbol{w} \propto S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$